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ORIGINAL PAPER

Motivational states in an undergraduate mathematics course: relations between facets of individual interest, task values, basic needs, and effort

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Abstract
According to situated expectancy-value models, motivational variables play an important role in students' engagement in academic mathematics courses and achievement. However, for academic mathematical learning, only a few studies provide information regarding the fluctuation of task values on different occasions (situation-specificity, state) and the extent to which task values depend on individual characteristics (personal-specificity, trait). Thus, answering these questions provides information about the extent to which adequate situational characteristics, such as relevance interventions, can influence students' motivation to engage in the learning process. To contribute to this discussion of the situation- and personal-specificity of task values, I surveyed 181 undergraduate students attending an advanced mathematics course. These students reported on their task values and effort three times in four different lessons during a course in linear algebra. I focused on the following three questions: (1) situation-specificity: to what extent do task values fluctuate between situations (and between persons)? (2) predictions of task values: to what extent do personal characteristics and situational perceptions predict task values? (3) predictions of effort: to what extent do personal characteristics and situational perceptions predict effort? These questions were answered by applying linear mixed models. The analysis results indicate that task values fluctuate only slightly in different situations and are strongly predicted by individual characteristics, such as interest in university mathematics, and task values predict effort. The theoretical and practical implications of the results are discussed.

Keywords Task values · Interest · Basic needs · Effort · Trait–state · University mathematics course

1 Introduction

Task value, defined as the value attributed to a task by students, is an important construct to explain students' choices and achievement in learning processes. Based on expectancy-value models, empirical results suggest that higher task values relate to more effort in higher education (Dietrich et al., 2017). In a recent contribution, Eccles and Wigfield (2020) claimed that situational aspects must be considered when analyzing such relationships between task values and indicators of successful learning processes. Furthermore, self-determination theory may give fruitful insights to link situational aspects and task values in mathematical learning processes. The self-determination theory proposes that the experiences of autonomy, competence, and social relatedness (known as the basic needs) affect motivation in learning processes (Deci & Ryan, 2002), and thus, the basic needs can serve as indicators of students' perceptions of situational aspects. Wigfield and Koenka (2020) questioned whether it is useful to integrate the basic needs in social-cognitive theories, which in this case implicates the expectancy-value theory, in order to develop the motivation field.

The entry phase of a university level mathematics study program was chosen as the study context because the first semester in a study program is crucial, as indicated by the high dropout rates (Geisler & Rolka, 2021; see also OECD, 2010). A reason behind students dropping out of mathematical programs is that they struggle to value the learning content, in particular mathematics as a scientific discipline (di Martino & Gregorio, 2019; Kosiol et al., 2019; Liebendörfer & Schukajlow, 2020). Additionally, the lecture, as a learning opportunity, in the entry phase of a study program is

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an unfamiliar learning opportunity for students (Gueudet, 2008).

Specifically, in this transition to a new learning culture and content, it seems plausible that value beliefs are situation-specific and dependent on students' perception of the learning situation. The existing literature does not shed enough light on how fluctuating task values are situated in the context of a university mathematics program, and how to support students to develop beneficial value beliefs in this context. As a first step in this direction, I addressed the first aspect and analyzed the extent to which situational and personal factors predict the emergence of task values and determine the ways in which task values predict effort. The results demonstrate how stable task values are or may be modifiable by relevance interventions. Moreover, this study contributes to the relationship between tasks values and effort, that is, I address the significance of students' evaluation of the learning content for their decision on how much energy and time they spend on the learning process. To analyze the possible predictors of task values and effort, I surveyed 181 students who rated their task values at three different time points in four lessons during a first-semester mathematics course.

2 Theoretical and empirical background

2.1 Learning mathematics at university

In Germany, similarly to many countries (Gueudet, 2008), there is a substantial gap between school and university regarding the teaching and learning of mathematics. Two major differences between these two educational settings play an important role in the transition from school to university mathematics: (1) a shift in the form of mathematics from a more applied form (at the school level) to mathematics as a more scientific discipline (at the university level), and (2) a change in the learning culture from a more guided form of learning to a more self-regulated one (Clark & Lovric, 2009; Rach & Heinze, 2017). The first aspect relates to the goals of education; one important educational goal of mathematics learning in schools is to help students apply mathematics for solving real-world problems (OECD, 2016); I call this special form of mathematics 'school mathematics'. Conversely, in 'university mathematics', mathematics is emphasized as a scientific discipline, introduced to learners in the entry phase of the study program by formal definitions of concepts and deductive proofs (Engelbrecht, 2010). As the practices for mathematical instruction differ between schools and universities, students are not familiar with the highly demanding practices at universities; consequently, they struggle to understand the concepts and may intend to drop out of the program in the first semester (Geisler & Rolka, 2021). Specifically, students enrolled in a teacher education program develop less interest in university mathematics as compared to mathematics students (Ufer et al., 2017), though they have many courses on university mathematics within their study program in Germany (Buchholtz, 2017). The second aspect, a shift in how mathematics is taught and learned, is visible in the organization of learning opportunities in the two educational settings. At school, teachers guide learners through a learning path, whereas in university, the focus of learning lies in the self-study phases, where students must complete highly demanding weekly tasks (an empirical analysis concerning these tasks in Germany can be found in Weber & Lindmeier, 2020). These self-study phases are accompanied by lectures, where a lecturer typically presents mathematical content, often in a definition-theorem-proof structure (Engelbrecht, 2010) and by tutorials in which the weekly tasks are prepared and discussed. After presenting the learning context in which the study is embedded, I address the central constructs of this study.

2.2 The role of task values in mathematical learning processes

2.2.1 Conceptualization of task values in situated expectancy-value models

Expectancy-value models are used to explain learners' behaviors in learning processes and connect individual expectations and values with the results of learning processes (Eccles & Wigfield, 2020). In this study, I concentrate on values that are defined as the "value (or valence) of an activity with respect to its importance to the individual" (Wigfield & Cambria, 2010, p. 3). Eccles and Wigfield (2020) used the term "task values" to focus on the value of a specific task that is ascribed by learners. They conceptualized task values as being subjective (Eccles & Wigfield, 2020, p. 4; see also Gaspard et al., 2015): "First, we argued that task values are subjective, meaning that the same task can be valued quite differently by different individuals and tasks with equivalent levels of difficulty can be valued quite differently by any one person." Thus, task values is not an invariant construct but is constructed by an individual in a certain situation, implying that the situation-specificity of task values must be considered. The construct of task values is multidimensional, consisting of the following four components: attainment value (how relevant it is for an individual to cope successfully with a task and how competent the individual is when dealing with the task), intrinsic or interest value (to what extent does the individual find dealing with the task enjoyable), utility value or usefulness of the task (to what extent is dealing with the task sensible for present and future aims), and costs (how much does the individual

have to invest to cope with the task) (Eccles, 1983; Gaspard et al., 2015). There are some studies on intrinsic value that is associated with positive emotions (Pekrun, 2006). Contrastingly, studies on attainment value that is closely associated with learners' identity (Gildehaus, 2021) are limited. Some authors, for example, Gaspard et al. (2015), developed sub-components of attainment value, such as personal importance and importance of achievement. In recent studies, utility value was also divided into sub-components—for example, in the work of Gaspard et al. (2015), which was embedded in the context of mathematics in school classrooms, utility value was differentiated into general and societal utility and utility for school, daily life, and jobs. With regard to costs, there is an emerging discussion on whether this is the third variable, after expectancies and values (Barron & Hulleman, 2015), or whether it is simply a component of values (Eccles & Wigfield, 2020).

Task values relate highly to situational interest (Hidi & Renninger, 2006; Krapp, 2002): the component 'intrinsic value' is conceptualized and operationalized similarly to the feeling-related component of situational interest; the components 'attainment value' and 'utility value' relate to the value-related component of situational interest (see also Linnenbrink-Garcia et al., 2013; Parrisius et al., 2021). Therefore, results concerning situational interest can be used to gain a partial but better insight into the role of task values in mathematical learning processes.

2.2.2 Intra- and inter-individual differences in task values

An intra-individual approach "investigates how the motivational experience of an individual differs between situations" (Dietrich et al., 2017, p. 54). This approach can complement inter-individual approaches commonly employed in the past, when using expectancy-value models (Eccles & Wigfield, 2020). However, few studies have analyzed intra-individual differences in motivational variables. Dietrich et al. (2017) conducted a study focusing on differences in task values depending on the learning topic or the specific learning situation. Across ten lessons of an educational psychology course (ten topics), three times a lesson (three situations), 155 pre-service teachers rated their expectancies, task values, costs, and efforts. Applying multilevel structural equation modeling, Dietrich et al. (2017) analyzed the extent to which expectancies, task values, and costs differed between situations, topics, and students. On every level—the specific learning situation, learning topic (that was different for every lesson), and students—the variability of task values was identified, that is, persons differ in their estimation of motivation between each other and in their motivational states between different topics or situations. Therefore, the authors concluded that task values depended on the specific situation and learning topic, which are both intra-individual factors; thus, measuring task values only on a domain level and as a personal characteristic overlooks important features of the construct. In contrast, Dietrich et al. (2017) did not analyze inter-individual differences in depth; thus, they did not identify traits that predict the emergence of motivational states.

An initial insight into the question of whether task values or, more generally, motivational characteristics fluctuate more between persons (inter-individual differences) or between situations (intra-individual differences), is provided by the following three studies. First, Tanaka and Murayama (2014) analyzed the stability of situation-specific interest in a university study program. A total of 158 undergraduate students reported their interest once after each lesson of an introductory psychology course, for a total of twelve lessons. The researchers identified a substantial amount of intra-person variability (70%) in interest over the course of the semester. Second, Tsai et al. (2008) analyzed the variance in seventh-grade students' interest experiences in three different subjects at school, namely, mathematics, German, and a second foreign language. During the study duration of three weeks, 261 secondary school students reported their interest in every subject. Using hierarchical linear modeling, the researchers showed that 36% of the variance in students' interest experiences in mathematics lessons was located at the intra-student level (similarly to the results of the second foreign language). Regarding German, the variance in the students' scores was slightly higher (44%). Third, a recent study by Parrisius et al. (2021) focused on the fluctuation of task values in ninth-grade mathematics classes. Using linear mixed modeling, Parrisius et al. (2021) found that intrinsic value and importance (a combined scale of attainment and utility value) varied substantially between and within students, whereas variances in intrinsic value were attributed more to fluctuation aspects, though importance depended mostly on variances between students, and thus on time-consistent aspects.

Combining the results of these empirical studies, one can assume that motivational variables, such as task values, fluctuate to a large extent between learning situations at the university level, but this fluctuation may be to a lower extent in courses with a focus on mathematics. Further evidence is needed to support these assumptions.

2.2.3 Personal and situational determinants of task values

The fluctuation of motivational states was the focus of the last section; this section summarizes the studies that address variables that are responsible for the systematic variation of motivational states. First, the personal characteristics that are assumed to predict the emergence of task values are summarized. Individual interest in learning content is often characterized as a personal trait that is assumed to influence value beliefs concerning a learning

situation. This is a logical assumption as the learning content is central to the situation. Empirical results confirm this assumption in different contexts, for example, for a tenth-grade social science class (Ferdinand, 2014), for a science class during a three-week residential summer program (Linnenbrink-Garcia et al., 2013), for a seventh-grade mathematics class (Tsai et al., 2008), and for a ninth-grade mathematics class (Schukajlow & Rakoczy, 2016). In contrast, the relationship between cognition-related variables and task values is an open-ended question. Schukajlow and Rakoczy (2016) did not find a relationship between prior knowledge and joy (related to intrinsic value) in a mathematical learning situation at school. Contrastingly, Rotgans and Schmidt (2011) reported a small but significant relationship between prior knowledge and situational interest of students in economics in a problem-based environment. Interpreting the study results and transferring them to the context of a university mathematics course, there is a need to consider whether the cognitive and motivational traits correspond to the requirements of the learning situation (see Ufer et al., 2017). As the learning content changes during the transition from school to university—from an applied form of mathematics to a scientific form of mathematics (see Sect. 2.1)—it is possible to differentiate interest in school and in university mathematics empirically.

Second, the interactions of persons and situations, in the sense of a person's perception of the situation, are considered. These perceptions are frequently used as mediators between features of the learning situation (e.g., openness of the lesson) and learners' outcomes (e.g., emergence of interest) (Hartinger, 2006). In the self-determination theory of Deci and Ryan (2002), the three basic needs (autonomy, competence, and relatedness) are related to the emergence of situational interest and motivation (e.g., Kiemer et al., 2015; not supported by Schukajlow & Krug, 2014). Specifically, experiences of competence have a mediocre relation with situational interest (Ferdinand, 2014, partially confirmed by Willems, 2011; see also Linnenbrink-Garcia et al., 2013); experiences of autonomy are shown to be related to situational interest, in particular, the element of finding the situation conducive to one's wishes (Ferdinand, 2014; Willems, 2011). Experiences of social inclusion were found to be less important as compared to those of autonomy and competence for task values (Ferdinand, 2014; Willems, 2011). One reason for these interrelations between experiences of competence, autonomy, and task values is that if the students' basic needs are fulfilled, the value-related features of the learning situation are perceived by the students, which enhances task values. Such features of the learning situation may be reflection tasks: students must write a letter to someone close to them to explain the meaning and utility of mathematics and refer to the content of the course. Such tasks may enhance the interest of students enrolled in a teacher education program (Liebendörfer & Schukajlow, 2020).

2.3 The role of effort in mathematical learning processes

In the present study, effort is regarded as an indicator of successful learning processes. It is differentiated from engagement in the sense that engagement is a multidimensional construct with emotional, cognitive, and behavioral aspects, whereas effort refers only to behavioral aspects and focuses on the time and energy invested by learners in their learning process (Patall et al., 2016).

Effort (regulation) is a variable partially used as a learning outcome (e.g., Pekrun, 2006 in control-value-theory; Song & Chung, 2020) or as a learning strategy (Liebendörfer et al., 2022). Although effort is analyzed from these two perspectives, only a few studies identify relations between effort (regulation) and personal traits or basic needs, e.g., relations between effort regulation and prior achievement (Liebendörfer et al., 2022; Schiefele et al., 2003) and between effort and study interest (Schiefele et al., 2003).

The following theoretical chain explains the proposed prediction of effort by task values: if learners valued the content highly, they would use deep learning strategies (Willems, 2011) and engage deeply in the learning process (Schiefele et al., 2003) to learn more about the content. In particular, task values and, to a lesser degree, expectations probably relate to effort, as indicated by the following researchers: Dietrich et al. (2017) in an educational psychology course; Song and Chung (2020) with data of the TIMS study of Korean ninth-grade students; and Guo et al. (2016) with German ninth-grade students. According to Guo et al. (2016), attainment value relates to effort. As teacher education students probably value the mathematical content less than mathematics students (Ufer et al., 2017) and as task values probably relate to effort, it seems plausible that teacher education students put less effort into dealing with the content than do mathematics students.

3 Research questions and hypotheses

Authors of previous research stated that students struggle to value mathematics as a scientific discipline. This struggle leads to less satisfaction with the study and ultimately increases the risk of dropout (Geisler & Rolka, 2021; Kosiol et al., 2019; Liebendörfer & Hochmuth, 2013). Constructing a suitable learning identity implies high and stable value beliefs to engage in an unknown learning culture. To the best of my knowledge, there are only a few studies concerning higher education courses in mathematics that deal with the

emergence of motivational variables, such as task values, and personal and situational factors that may be related to this emergence (e.g., Liebendörfer & Schukajlow, 2020). Theoretical frameworks indicate that there are bidirectional effects between task values, effort, and experiences of autonomy and competence. In this study, the emergence of task values and effort is in the center, and hence, I analyze one direction of this relationship, namely, that in which task values and effort are the dependent variables. This one-directional relationship is indicated by the one-way arrows in Fig. 1.

Changes in task values and effort may be due to changes in the characteristics of the learning situation, such as the learning topic, the teaching method, etc., or factors that are indirectly dependent on the learning situation, such as students' fatigue; therefore, intra-individual approaches are useful in this context. Additionally, previous research (e.g., Schukajlow & Rakoczy, 2016) shows inter-individual differences in motivational states, which are also discussed in this study.

I conducted this study in a typical first-semester lecture course titled "Linear Algebra", where students rated their task values and effort three times in four lessons. I focused on the following three research questions that shed light on the role of task values in university mathematics learning processes:

1. Situation-specificity of task values: to what extent do task values fluctuate between situations (and between persons)?

   I assume that students in this university mathematics course differ in their task values to a large degree. Since each lesson focuses on a different mathematical topic, I also expect that task values fluctuate substantially between the times of lessons in the semester. Concerning situation-specificity in one lesson, I have no clear assumption because, on the one hand, the teaching method and the topic do not fluctuate substantially in one lesson, on the other hand, the learning situation is unique, and as the lessons are 90 minutes long, the fatigue effect may influence the ratings of value beliefs (H1, construct 'task values' in Fig. 1).

2. Predictions of task values by personal characteristics and situational perceptions: to what extent do personal characteristics (study program, prior achievement, and interest facets) and situational perceptions (experiences of autonomy and competence) predict task values?

   I assume that students enrolled in a teacher education program report lower task values than students enrolled in a mathematics program (Ufer et al., 2017*). Additionally, students with a higher interest in university mathematics show much higher value beliefs than students who are less interested in university mathematics. As school mathematics is not the learning content in the entry phase of the university mathematics course, I have no clear assumption concerning the role of interest in school mathematics for task values when controlling for interest in university mathematics. As students who performed well in school have probably a higher ability to understand university mathematics* (Rach & Heinze, 2017*), they may report a higher level of task values than other learners. Thus, prior achievement may also predict task values* (H2.1, short arrow left in Fig. 1). *I assume that situational perceptions predict task values. Feeling autonomous means that the learn-*

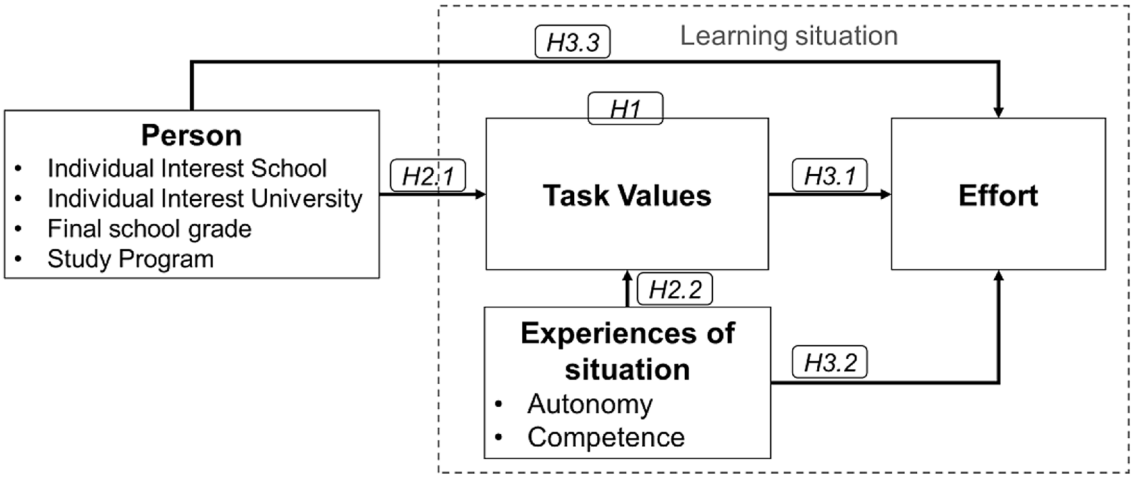

Fig. 1 Model proposed in this study

ing situation matches one's wishes, leading to valuing the situation highly, that is, higher task values. When students feel competent in a learning situation, they are not overburdened, unlike the case in the entry phase in a university mathematics course, and consequently they value the learning content at a higher level than students who feel less competent. According to the empirical results of Ferdinand (2014) *and* Willems (2011)*, I expect that experiences of autonomy predict task values to a higher level than experiences of competence* (H2.2, arrow on the bottom right in Fig. 1).

3. Predictions regarding effort by personal characteristics and situational perceptions: to what extent do personal characteristics (study program, prior achievement, and interest facets) and situational perceptions (task values, experiences of autonomy and competence) predict effort?

I expect that students' perceptions of the situation predict effort; students who strongly value the learning content and tasks will engage in the tasks (H3.1, arrow right in Fig. 1). I assume the same to be true for students' feeling themselves to be autonomous and competent learners (H3.2, arrow on the bottom right in Fig. 1). In contrast to state variables, I assume that personal characteristics do not predict effort, except for students' prior achievement, indicated by the final school grade (see Dietrich et al., 2017*); consequently, I assume that students with a better final school grade report more effort because the final school grade is an indicator of cognitive resources and of the willingness to learn* (Trapmann et al., 2007) *(H3.3, arrow top in Fig. 1).*

4 Method

4.1 Sample

The sample comprised 181 students of an undergraduate mathematics program at a German university (age $M = 19.6$ years, $SD = 2.0$ years, $n = 150$; 94 male, 56 female, 31 no information) and a majority of them ($n = 131$) were in their first semester. Data on age, gender, and study program were missing for some students, who were not present at the first measurement.

Approximately 5% of the students in the lecture did not consent to participate in the study. The remaining students voluntarily participated in the study, and informed consent was obtained. I told them that their motivation during lessons was analyzed to improve the first year of their study program. The participating students were enrolled in different undergraduate programs: 108 students were in a teacher education program for upper secondary schools, 20 in a mathematics program, and 22 in other science, technology, and engineering (STE) programs, such as physics, computer sciences, etc. I have no information about the study program of the remaining 31 students. By including a mathematics course in teacher education programs for upper secondary schools, students should be introduced to mathematics as a scientific discipline and earn a substantial amount of content knowledge in mathematics. In the first academic year at many German universities, the undergraduate study programs for teacher education and mathematics students are similar and students from STE programs often join the same mathematics courses.

4.2 Design

This study is part of the project 'Situational Interest in a Mathematics study program' (SIMs). In the report by Rach (2020), the first results of this study that deals with the relationships between task values and basic needs can be found; the reported results were based on the data of one lesson only. The lessons in the course 'Linear Algebra' were structured as typical mathematical lectures, with a focus on formal definitions, mathematical statements, and deductive proofs. The course consisted of 14 weekly lectures from October to January, with each lecture lasting 90 min. In four lessons (conducted in October, November, December, and January), the students were asked to report their task values, experiences of competence and autonomy, and effort, in paper–pencil questionnaires at three time points: after 15 min, 45 min, and 75 min of the lesson (see Fig. 2). The exact time points of the measurement lay in intervals of five minutes around minutes 15, 45, and 75 of the lessons. A scientific member of a mathematics education group organized the measurement and gave the lecturer a sign when it was time for students to fill in the questionnaire. Then, the lecturer paused the lesson for three minutes and the students filled in the questionnaire. Some sweets were given as incentives to the students in each of the four lessons. The lecturer was told to conduct the four lessons as usual and no hints that the lessons were special in their content or implementation were provided.

In the first lesson, students also completed a questionnaire containing items that investigated their background information, such as their study program, final school grade, and individual interest facets (see Fig. 2); these variables are considered as traits in this study.

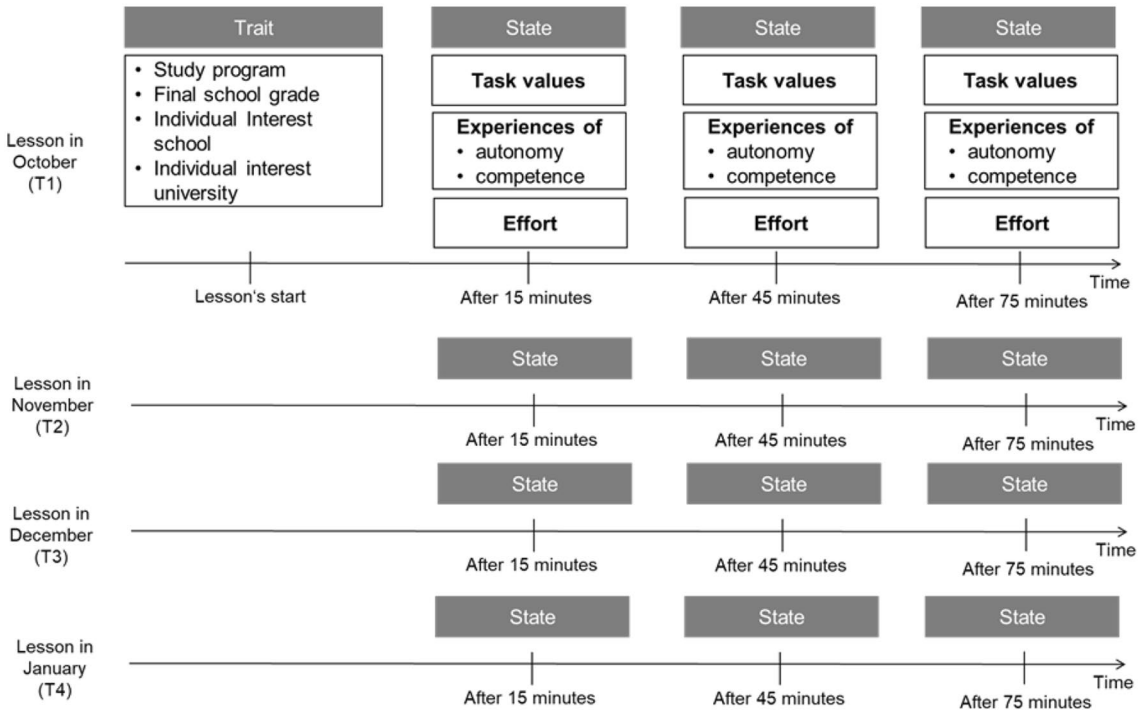

Fig. 2 Design of the study

At first glance, the form of teaching and the type of content in the four lessons in which the study was conducted were the same. This included the lecturer explaining the mathematical learning content and recording definitions, theorems, and proofs on a blackboard. However, in the in-depth observation of the lessons, it was perceived that the lessons changed with respect to the specific content that the students had to learn: in October, the lecturer presented important statements concerning injective and surjective functions, whereas, in November and December, fields and matrices were in the focus. In January, the mathematical content discussed until that point was applied to error correction codes. It is natural that the learning content changes from lesson to lesson but the changes in these lessons also influenced how the lecturer presented the content to the students: in the first and fourth lessons, the lecturer highlighted the importance of the presented content for mathematical and realistic issues; in the second lesson, he asked students to reflect on some mathematical questions, mainly in the first and second parts of the lesson.

In a university mathematics program, there is often a high dropout rate in the first semester (Geisler & Rolka, 2021). Therefore, students who dropped out of their study program or the specific course could not participate in measurement points at a later stage. This may be one of the reasons behind why fewer students participated in the study at the later measurement points: $n(\mathrm{T1}) = 150$, $n(\mathrm{T2}) = 113$, $n(\mathrm{T3}) = 95$, and $n(\mathrm{T4}) = 72$. Hypotheses H1, H2.2, H3.1, and H3.2 were tested by using data from 181 students, whereas for testing H2.1 and H3.3, information on students' background information was required, and thus, only the data of 150 students could be used.

4.3 Measures

To measure *state variables*, students were asked to consider the content of the past few minutes of the lesson and to complete the questionnaire within 3 min. In this study, I used questionnaires adapted from Dietrich et al. (2017), Linnenbrink-Garcia et al. (2013), and Willems (2011). Experiences of competence (item: "I have the feeling that I can understand difficult content"), experiences of autonomy (item: "I have the feeling that the lecture is as I wish"), and task-related effort (item: "I engage") were measured with single items, and task values were measured by a scale including four items that represented different components of task values: "I like these contents" (intrinsic value), "It is important for me to know a lot about the contents" (attainment value), "The contents are important for my study" (utility value for study), and "The contents are important for my professional life" (utility value for a job). In this study, I used two items for utility value: one for the near future (utility value for study) and one for the far future (utility value for a job). This is because teacher education students are the focus of this study and these students probably report specific motivation concerning their university courses and jobs. According to

the results in the field of expectancy-value theory (Dietrich et al., 2017), I did not integrate costs in the construct of task values. All items were approved with 59 students in a course of Didactics of Geometry beforehand. Descriptive data concerning task values for every measurement point can be found in Table 2, where it can be observed that internal consistency ranged from adequate to good and no evidence of either ceiling or floor effects were discovered.

To measure *trait variables*, I applied questionnaires developed and validated in other projects. In the project 'Self-concept and Interest when Studying Mathematics' (SISMa), Ufer et al. (2017) developed an instrument that differentiated between individual interest in school and in university mathematics. A sample item for interest in school mathematics was, "I am interested in the kind of mathematics that I learned at school." For university mathematics, they used similar items but replaced the word "school" with "university". Students rated all items for state and trait variables on a 4-point Likert scale ranging from 1 (disagree) to 4 (agree). Mean scores for all scales were computed on the valid responses under the condition that more than half the items had a valid response. Students also reported their study program (see Sect. 4.1) and their final school qualification grade from 1.0 (very good) to 4.0 (sufficient), which is a well-known indicator of prior achievement (Trapmann et al., 2007). Descriptive analyses and intercorrelations among the trait variables are presented in Table 1 and indicate nearly no interrelations between the trait variables. Attention should be paid to the strong relationship between interest in university mathematics and the study program when analyzing the prediction of task values and effort by personal traits.

4.4 Analysis strategy

The state values depended on the person and time point of the lecture (T1, T2, T3, and T4) and during the lecture (beginning, middle, and end). Therefore, linear mixed models were used to estimate the situation-specificity of task values and identify factors that predicted task values and effort. A random effect considers the variability of situations

Table 1 Sample size (*N*), mean (*M*), standard deviation (*SD*), internal consistency (Cronbach's α), and intercorrelation among the trait variables

Scale	*N*	*M* (*SD*) overall	α	IS	IU	*M* (*SD*) teacher education program	*M* (*SD*) mathematics program	*M* (*SD*) STE program
Interest in school mathematics (IS) (5 items)	147	2.90 (0.55)	.74	–		2.93 (0.53)	2.87 (0.58)	2.77 (0.63)
Interest in university mathematics (IU) (5 items)	149	2.57 (0.59)	.81	.18*	–	2.42 (0.54)	3.05 (0.49)	2.83 (0.63)
Overall school grade (1 item)	141	2.35 (0.61)		− .13	− .07	2.37 (0.60)	2.42 (0.72)	2.15 (0.51)

Items for interest facets were rated on a 4-point Likert scale ranging from *disagree* (1) to *agree* (4); overall school grade ranging from *very good* (1) to *sufficient* (4), *$p < .05$

Table 2 Descriptive statistics (*M*—mean, *SD*—standard deviation), internal consistency (Cronbach's α), and intercorrelation among the study variables

	M	*SD*	α	TV11	TV12	TV13	TV21	TV22	TV23	TV31	TV32	TV33	TV41	TV42
TV11	2.89	0.56	.63	–										
TV12	2.82	0.60	.66	.84	–									
TV13	2.89	0.63	.69	.66	.73	–								
TV21	2.56	0.63	.70	.55	.62	.57	–							
TV22	2.71	0.66	.73	.60	.63	.56	.75	–						
TV23	2.65	0.70	.78	.62	.65	.60	.77	.90	–					
TV31	2.63	0.53	.60	.45	.43	.46	.61	.58	.61	–				
TV32	2.67	0.58	.69	.46	.50	.45	.56	.62	.56	.65	–			
TV33	2.56	0.64	.77	.48	.50	.45	.61	.66	.65	.64	.83	–		
TV41	2.71	0.58	.69	.41	.41	.41	.58	.51	.52	.56	.64	.64	–	
TV42	2.63	0.63	.79	.50	.48	.49	.59	.69	.69	.49	.73	.76	.66	–
TV43	2.47	0.76	.83	.40	.41	.33	.41	.46	.58	.32	.45	.54	.60	.69

TV11 = task values session 1, time 1 (after 15 min), TV12 = task values session 1, time 2 (after 45 min), TV13 = task values session 1, time 3 (after 75 min), TV21 to TV43 for sessions 2, 3, and 4, respectively. Items were rated on a 4-point Likert scale ranging from *disagree* (1) to *agree* (4); all correlations were significant at $p < .05$

in which the persons stated their task values. These effects give variation to the measurement, but the effects are not the focus of the research questions, except in research question 1, which deals with the variation of task values between situations. For computing random effects, I used the following variables: the person and time point of and during the lecture. Variables, where all levels of interest are included in the study, are used as fixed factors (Magezi, 2015). To compute fixed effects, I used the following variables: personal traits, such as individual interest, prior achievement, study program, or situational states, such as experiences of autonomy and competence. I computed all analyses in R (version 4.1.0), using the package lme4 (Bates et al., 2015), and reported these results. As the dependent variable 'effort' is measured with one item at each measurement point, I also conducted the analyses concerning this variable with the package ordinal (Haubo, 2019) and obtained results similar to the results obtained from analyses with the package lme4.

The marginal R-squared considers only the variance of the fixed effects, while the conditional R-squared considers both fixed and random effects (possibilities for calculating these two types of R^2 are shown in Nakagawa & Schielzeth, 2013). Linear mixed models have the advantage of handling cases with missing values (Hilbert et al., 2019). Thus, cases with missing values are still involved in the analysis; due to the design of my study, an average of 41% of the data were missing per variable. When interpreting the different models, one must keep in mind that the models are based on different amounts of observations.

5 Results

First, I provide a descriptive overview of the data. In Table 2, descriptive statistics, internal consistency, and intercorrelation among task values at different time points are shown. It can be observed that task values across and within each session were either moderately or highly correlated, with the largest correlations being within each session.

5.1 Variability of task values in situations

To analyze the extent to which the variance of task values is dependent on a person, on the time point of the lesson (T1, T2, T3, T4), and on the time point during the lesson (beginning, middle, and end), I conducted a linear mixed model with these factors as random effects. As a dependent variable, I used the measures of task values for every student at every time point during a lesson and every chosen lesson. The results are presented in Table 3.

A large extent of variance in task values is dependent on the person. There is little variance between the sessions during the whole semester and no variance in one session; thus, H1 is only partly confirmed.

Table 3 Explanation of variance of task values

Random effects	Variance	*SD*
Person	0.268	0.518
Time point of the lecture	0.029	0.172
Time point during the lecture	0.000	0.011

1284 observations, 181 persons, four lessons during the semester, three time points during lessons

5.2 Prediction of task values by personal characteristics and experiences of autonomy and competence

The analysis concerning research question 1 indicates that while task values fluctuate between persons and situations, the degree of fluctuation is less in the latter case (Model 1, Table 4, see also Sect. 5.1). Thus, in H2.1, I assumed that personal traits, such as the chosen study program, prior achievement indicated by the final school grade, and individual interest facets, probably predict task values. The results of Model 2, shown in Table 4, partially support H2.1: the chosen study program and individual interest in university mathematics predict task values, whereas there is no empirical support for the prediction of task values by prior achievement and interest in school mathematics. Specifically, students enrolled in the teacher education program reported lower scores of value beliefs than the scores reported by students enrolled in the mathematics program (see also Table 6 in the Appendix). The regression coefficient $b = 0.36$ indicates that students with a higher individual interest in university mathematics reported higher task values. Including the trait variables in the model, $R^2_{marginal} = 28\%$ of the variance in task values can be explained by these variables.

In Model 3, I analyzed how other state variables, such as experiences of autonomy and competence, predict task values in a specific situation. The results show that while experiences of autonomy strongly predict task values, the prediction by experiences of competence is weaker in comparison, confirming H2.2.

When combining Models 2 and 3 in Model 4, I did not integrate the variable *time point during the lecture* in the model because there is no variance in task values regarding this variable (see Sect. 5.1). Model 4 shows that the chosen study program, individual interest in university mathematics, experiences of autonomy, and experiences of competence, explained 40% of the variance in task values in different learning situations.

Table 4 Prediction of task values by personal traits and states

Fixed effects	Model 1		Model 2			Model 3			Model 4		
			Estimate	*SE*	*t*	*Estimate*	*SE*	*t*	*Estimate*	*SE*	*t*
Intercept			1.60	0.27	5.95***	1.60	0.11	15.24***	1.18	0.17	7.07***
Personal traits											
Study program 1			0.43	0.10	4.45***				0.32	0.09	3.51***
Study program 2			0.47	0.10	4.68***				0.32	0.09	3.63***
Prior achievement			− 0.02	0.05	− 0.37						
Interest University			0.36	0.06	5.93***				0.17	0.06	2.97**
Interest School			0.00	0.06	0.01						
States											
Autonomy						0.28	0.02	14.60***	0.25	0.02	12.46***
Competence						0.13	0.02	7.01***	0.11	0.02	5.72***
Random effects	*Var*	*SD*	*Var*	*SD*		*Var*	*SD*		*Var*	*SD*	
Person	0.268	0.518	0.123	0.351		0.157	0.396		0.112	0.335	
Time point of the lecture	0.029	0.172	0.030	0.174		0.029	0.170		0.031	0.177	
Time point during the lecture	0.000	0.011	0.000	0.019		0.000	0.007		–	–	
Observations	1284		1101			1267			1117		

Model 1: $R^2_{conditional} = .66$; Model 2: $R^2_{conditional} = .64$, $R^2_{marginal} = .28$; Model 3: $R^2_{conditional} = .70$, $R^2_{marginal} = .25$, Model 4: $R^2_{conditional} = .73$, $R^2_{marginal} = .40$ (for computing model 4, I did not integrate the time point during the lecture because of no variance in task values); *Var*: variance, *SE*: standard error, *SD*: standard deviation. Study program 1: difference between teacher education students and mathematics students; Study program 2: difference between teacher education students and STE students; levels of significance: $^{**}p < .01$, $^{***}p < .001$

5.3 Prediction of effort by task values, experiences of autonomy and competence, and personal characteristics

To predict students' effort in learning processes, I conducted the following analyses with state and trait variables (see Table 5). Model 1 considers task values as a state variable that predicts 5% of the variance in effort (supporting H3.1). Controlling task values, the other state variables, experiences of autonomy and competence, do not have additional effects on effort (Model 2; does not support H3.2). Inserting trait variables in the model (H3.3), there is only one single trait variable that explains variance in effort, and this variable is the chosen study program: teacher education students reported more effort than STE students. Even if the study program is deleted from the list of predictors, no other personal trait predicts effort. Integrating task values and the chosen study program in one model (Model 4), 7% of the variance in effort can be explained by these variables. The study program predicts effort, in the sense that students enrolled in the teacher education program report more effort than students enrolled in a mathematics program or STE program when controlling for task values (see also Table 6 in the appendix).

Noticeably, while there is only a slight variation in effort between lessons or time points in the lessons, there are large differences in effort between persons. Although there are large differences in effort between persons, the considered traits can only partially explain why persons differ in their effort.

6 Discussion

The variable *task values* is an important variable in students' learning processes. According to situated expectancy-value models, it influences learners' choices and success (see Eccles & Wigfield, 2020). Previous studies focusing on task values in a specific situation answered questions regarding the way people differ in their task values and the extent to which their value beliefs coincide with choices and success. In the present study, this approach, focusing on inter-personal differences in task values, was combined with an intra-personal approach, where task values of one person were measured several times and the fluctuation of task values was considered.

6.1 Relations between individual interest, task values, basic needs, and effort

The study was conducted in a linear algebra course for first semester students. Students in this course learn to understand and consider mathematics as a scientific discipline

Table 5 Prediction of effort by personal traits and states

Fixed effects	Model 1			Model 2			Model 3			Model 4		
	Estimate	*SE*	*t*	*Estimate*	*SE*	*t*	*Estimate*	*SE*	*t*	*Estimate*	*SE*	*t*
Intercept	2.41	0.13	18.93***	2.32	0.13	17.27***	3.11	0.41	7.55***	2.43	0.13	18.67***
Personal traits												
Study program 1							− 0.16	0.15	− 1.08	− 0.31	0.14	− 2.25*
Study program 2							− 0.32	0.15	− 2.07*	− 0.43	0.14	− 3.00**
Prior achievement							− 0.01	0.09	− 0.07			
Interest University							0.10	0.10	1.02			
Interest School							− 0.04	0.10	− 0.40			
States												
Task values	0.28	0.04	7.99***	0.25	0.04	6.20***				0.33	0.04	8.61***
Autonomy				0.02	0.03	0.81						
Competence				0.04	0.03	1.56						
Random effects	*Var*	*SD*		*Var*	*SD*		*Var*	*SD*		*Var*	*SD*	
Person	0.369	0.608		0.391	0.626		0.336	0.580		0.327	0.572	
Time point of the lecture	0.008	0.090		0.010	0.099		0.018	0.135		0.006	0.076	
Time point during the lecture	0.009	0.097		0.009	0.093		0.012	0.110		0.010	0.102	
Number of observations	1281			1264			1098			1131		

Model 1: $R^2_{conditional} = .63, R^2_{marginal} = .05$; model 2: $R^2_{conditional} = .65$, $R^2_{marginal} = .05$; model 3: $R^2_{conditional} = .60$, $R^2_{marginal} = .02$; model 4: $R^2_{conditional} = .62$, $R^2_{marginal} = .07$; *Var*: variance, *SE*: standard error, *SD*: standard deviation. Study program 1: difference between teacher education students and mathematics students; study program 2: difference between teacher education students and STE students; levels of significance: ***$p < .001$, **$p < .01$, *$p < .05$

(Rach & Heinze, 2017), which was valued at a low level, especially by many students enrolled in a teacher education program (Liebendörfer & Hochmuth, 2013). This observation is also replicated in this study.

Although I measured task values in four lessons and at three time points in each lesson, there was little fluctuation in the task values of one learner. My results indicate that value beliefs in this context have the character of a slowly modifiable construct more than one that changes from one occasion to another. This was surprising and differed from the results of Dietrich et al. (2017) because the four lessons focused on different topics, with corresponding changes in the teaching method; however, these changes might only slightly influence subjective task values. Beneath the mentioned methodological issues in the limitation, these results indicate that students' beliefs regarding the importance of the learning content are stable. This may be due to the phenomenon that the learners had no time to reflect on the value of the learning content because they could hardly follow the highly demanding lessons (di Martino & Gregorio, 2019).

Contrary to the low fluctuation between situations, task values between persons differed substantially in this study, which is partially explained by the chosen study program and individual interest in university mathematics. Students interested in university mathematics have a substantially better chance to develop task values in mathematics courses. Experiences of autonomy and competence are sensible factors to explain systematic variation in students' beliefs (Ferdinand, 2014). These relationships, empirically indicated, contribute to combine the expectancy-value and self-determination theories, in particular, task values with basic needs (Koenka, 2020). Whereas Hattie et al. (2020) proposed a close relationship between the need for competence and expectancy of success, this close relationship may be also true for the need for autonomy and task values.

To gain a better insight into the link between task values and successful learning processes, I also investigated the prediction of effort by task values. Effort is predicted by task values and the study program to a small degree (see Dietrich et al., 2017); specifically, teacher education students reported more effort than students of other programs. However, the variance in effort between students was not well explained by these variables. The factor of fulfillment of the basic needs also had no additional power to explain differences in effort. Thus, effort seems to be a variable that is not well connected to other state measures, such as task values or experiences of autonomy. Therefore, in this

study, I was unable to clarify why students of university mathematics courses differed in their effort. As effort seems to be an important factor in successful learning processes, predictors of effort should be addressed in future studies. The interplay between task values and expectancies may play a substantial role in predicting effort in this context, although this assumption was not validated with ninth-grade students (Song & Chung, 2020).

The study program variable has an ambiguous character in this study; while teacher education students show lower task values than mathematics students, they report more effort when controlling task values. Therefore, teacher education students are not disengaged in the learning content, but value the learning content less than other participants. Therefore, specific interventions for this group are needed to support learning motivation.

6.2 Limitations

The study design posed limitations for the interpretation. As I did not want to disturb the course substantially, I measured task values only three times in each of the four lessons. Some students did not attend lessons in the later phase of the semester, consequently, the sample size decreased at later time points (see Sect. 4.2). As the learning situations in which the students reported their task values belonged to one mathematics course, there were similar teaching patterns in every lesson. While reading the results, one must keep in mind that with this cross-sectional, longitudinal design, no causal relations can be drawn and the relations between the states are only correlative. Further, experimental designs are necessary to validate the assumed influences of one variable on the other.

As I focused on a large sample, I employed questionnaires for measuring trait and state variables. Specifically, the questionnaire for measuring state variables, such as task values and effort, contained only a few items. This approach of using short questionnaires was chosen to avoid disturbing the students and instructor during lessons, to avoid wearying students with similar items, and to consider ecological validity (see Tanaka & Murayama, 2014). In future studies, it may be worthwhile to differentiate task values into components, such as intrinsic, attainment, and utility values, to gain a deeper insight into students' value beliefs (see Parrisius et al., 2021). When analyzing different components of task values, it is necessary to use more items to build up the whole complexity of the different components and to avoid using only one item of one sub-component—in this study, I measured attainment value with one item of the sub-component 'personal importance'. Specifically, for students enrolled in a teacher education program, the distinction between utility value for their study program and utility value for their further job (in school) may contribute to identifying situations that are appraised by students as more valuable than other situations.

6.3 Theoretical and practical implications

The results of this study can extend the role of task values in two directions: first, the results show that in an undergraduate mathematics course, the fluctuation in task values from situation to situation, in this case from topic (e.g., injective functions) to topic (e.g., matrices) was not as large as expected. The fact that all topics belong to the field 'Linear Algebra' may explain this result and it could be of interest to analyze students' task values in different fields of mathematics, e.g., in Linear Algebra and Analysis. Also, this result may be specific to the mathematics domain because the results of Tsai et al. (2008) concerning fluctuation in motivation between German, a second foreign language, and mathematics also support this assumption. Noticeably, inter-individual differences in task values are predicted by personal traits, such as interest in university mathematics, in contrast to interest in school mathematics, which supports the need to differentiate mathematical interest in facets (Ufer et al., 2017). Second, the results provide insights into the engagement of teacher education students in university mathematics compared to the engagement of other science, technology, engineering, and mathematics (STEM) students. Although all students participated in the same lecture and mathematics is an essential part of their study program, teacher education students valued the content to a lesser extent than other students; however, they reported more effort. The result that students from a teacher education program value the content less than students from other study programs may be because the mathematical content does not take into account the specificity of the professional development of future teachers. Teacher education students may value the content more highly if it serves to build up specialized content knowledge (Ball et al., 2008) which is specific knowledge for teaching (see also the knowledge type "school-related content knowledge", developed by Dreher et al., 2018). The results of this study lead to the assumption that the relationship between value beliefs and effort is only valid for students from subject-oriented STEM programs, and teacher education students put effort into their learning processes for reasons other than value beliefs. As task values differed significantly between people and groups, analyzing task values with an inter-individual approach puts forth new ideas.

On the one hand, as task values do not depend greatly on the specific situation, one could argue that it is useless to implement value-supporting elements in the instruction. On the other hand, variance in task values was observed

between situations. In particular, for teacher education students, there is a need to design elements to enhance their value beliefs concerning the university mathematics content. As these students intend to become schoolteachers, they are often confused regarding the purpose of including university mathematics in their program curriculum (Geisler, 2018). To fulfill their wishes and not lose sight of the aims of university mathematics courses, mathematical tasks that link school mathematics and university mathematics could be implemented (Bauer & Kuennen, 2016; Weber et al., 2020). Recently, Liebendörfer and Schukajlow (2020) reported on relevance interventions for teacher education students in a lower secondary school program that may also have the power to enhance task values. In their study, they found no direct effect of their interventions on students' interest (in contrast to the results of Gaspard et al., 2015); however, the quality of students' reflections in the intervention related to students' interest at the end of the intervention. Combining their results with the results of the present study, it can be questioned whether short interventions are always the adequate way for enhancing students' value beliefs in this university context. Long-term interventions, in the sense that students get the opportunity to reflect several times during their learning process on the relevance of the content (Hulleman & Harackiewicz, 2009), seems to be necessary. In addition to implementing relevance interventions regularly in the lessons, it is important to enable students to identify these elements so that the elements can take their effect on students' motivation (Hulleman et al., 2017; Liebendörfer & Schukajlow, 2020). Basic needs, specifically, experiences of autonomy and competence, relate positively to task values. Therefore, if students do not feel autonomous or competent, it is difficult for them to value the mathematical content in university courses. Further research should analyze in specific contexts under which conditions, e.g., short- or long-term, interventions can enhance students' value beliefs.

Appendix

See Table 6.

Table 6 Descriptive statistics (*M*: mean, *SD*: standard deviations, *N*: number observations) of task values and effort, depending on the study program

Program	Task values			Effort		
	M	*SD*	*N*	*M*	*SD*	*N*
Teacher education mathematics	2.52	0.58	795	3.24	0.76	795
Mathematics	3.18	0.53	147	3.33	0.69	147
STE	3.11	0.48	189	2.95	0.89	189

Items were rated on a 4-point Likert scale ranging from *disagree* (1) to *agree* (4)

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Declarations

Conflict of interest The author declares no conflicts of interest relevant to the content of this article.

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